Test of November 27th, 2024

Physics Course at the Department of Computer Science
University of Milan

The first three exercises, if solved correctly, provide 18/30 points and are considered a prerequisite for accessing the correction of Part Two. If even one of these exercises is incorrect, Part Two will not be graded. Minor imperfections or small calculation errors will be tolerated.

Part 1: Basic Exercises

Exercise 1: Uniform Rectilinear Motion

A cyclist travels along a straight road with a constant speed of 5 m/s.

1. How long does it take to cover a distance of 200 m?
2. Draw the graph of the position as a function of time for the first 10 s.
3. If the cyclist suddenly stops after 40 s, what is the total distance traveled?

(Provide results in SI units.)

Exercise 2: Inclined Plane Without Friction

A block with a mass of $m = 10\,\text{kg}$ slides down a frictionless inclined plane that forms an angle of $30°$ with the horizontal.

1. What is the acceleration of the block along the inclined plane?
2. If the block starts from rest, what is its velocity after 5 s?
3. How far does it travel during these 5 s?

(Use $g = 9.8\,m/s^2$ for calculations.)

Exercise 3: Static Friction Force

A piece of furniture with a mass of 50 kg rests on a horizontal floor. The coefficient of static friction between the furniture and the floor is $\mu_s = 0.4$.

1. What is the maximum value of the static friction force that can resist the motion of the furniture?
2. If a horizontal force of 180 N is applied to the furniture, will it start to move? Justify your answer.
3. If the applied force is 220 N, calculate the initial acceleration of the furniture, given that the coefficient of kinetic friction is $\mu_k = 0.3$.

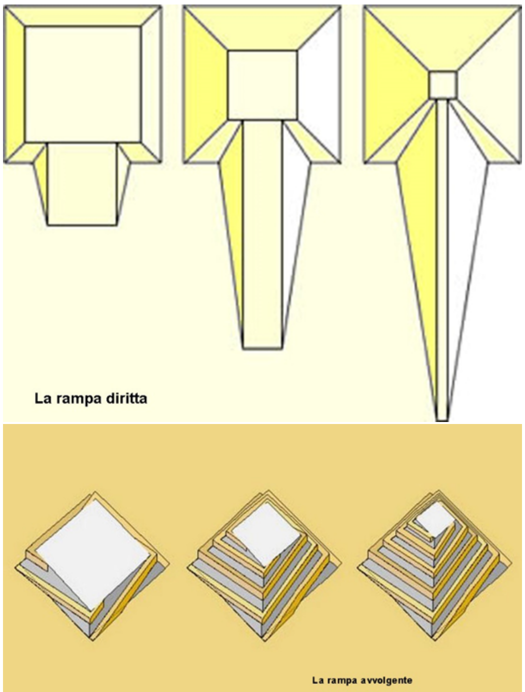
Part 2: Fermi Problem

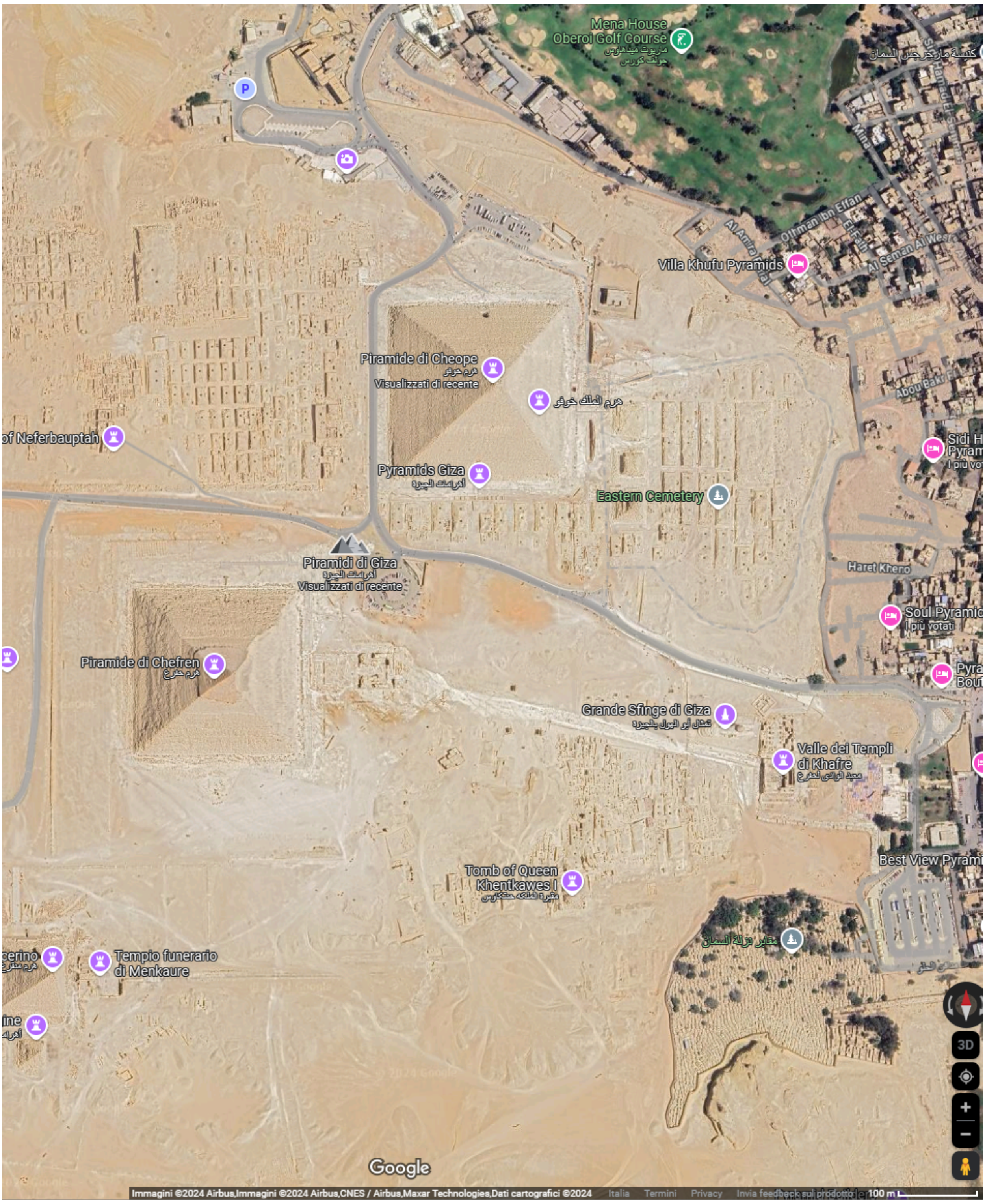
The Great Pyramid of Giza is one of the most impressive monuments of antiquity. It is estimated that its construction took approximately **20 years**. Does this seem like a reasonable time frame to you? Estimate the construction time based on the attached materials (considering that the coefficients of static and dynamic friction for dry sand are approximately $\mu_s = 0.5$ and $\mu_k = 0.4$, or for wet sand $\mu_s = 0.25$ and $\mu_k = 0.15$). Estimate forces, time, manpower, and the technologies used (ramps or other means). Formulate additional questions and apply what you have learned in class to answer them.

Quarry	**Distance from Giza (km)**	**Cumulative Elevation (m)**	**Usage Percentage (%)**	**Specific Weight (kg/m^3)**
Giza Plateau	0.5-1	10-20	80-85%	2200-2500
Tura	15	~20	10-15%	2600
Mokattam Hills	20	~30	<5%	2700
Aswan	~900	~100	<1%	3000

Table 1: Data related to the quarries used for the construction of the Giza pyramids, including the specific weights of materials.


(Ph. propria)
DIMENSIONE DEI BLOCCHI DELLA PIRAMIDE DI CHEOPE


La rampa diritta
La rampa avvolgente


Mena House
Oberoi Golf Course
Villa Khufu Pyramids
Piramide di Cheope
Visualizzati di recente
of Neferbauptah
Pyramids Giza
Eastern Cemetery
Piramidi di Giza
Visualizzati di recente
Haret Khenó
Piramide di Chefren
Grande Sfinge di Giza
Valle dei Templi
di Khafre
Tomb of Queen
Khentkawes I
Best View Pyrami
Tempio funerario
di Menkaure
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